

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS / R

M2 Practical No: 8

Aim: Performing two-way ANOVA using aov() (R).

Code:

```
library(tidyverse)
```

```
df <- read.csv("kepler-exoplanet.csv") %>%  
  select(koi_fpflag_nt, koi_period) %>%  
  filter(!is.na(koi_period))
```

```
grand_mean <- mean(df$koi_period)  
N <- nrow(df)
```

```
stats_table <- df %>%  
  group_by(koi_fpflag_nt) %>%  
  summarise(  
    group_mean = mean(koi_period),  
    n = n(),  
    group_var = var(koi_period)  
  ) %>%  
  mutate(  
    # SSB calculation:  $n * (\text{Group Mean} - \text{Grand Mean})^2$   
    ssb_part = n * (group_mean - grand_mean)^2,  
    # SSW calculation:  $(n - 1) * \text{Variance}$   
    ssw_part = (n - 1) * ifelse(is.na(group_var), 0, group_var)  
  )
```

```
SSB <- sum(stats_table$ssb_part)  
SSW <- sum(stats_table$ssw_part)
```

```
df_between <- nrow(stats_table) - 1  
df_within <- N - nrow(stats_table)
```

```
MSB <- SSB / df_between  
MSW <- SSW / df_within
```

```
f_statistic <- MSB / MSW  
p_value <- pf(f_statistic, df_between, df_within, lower.tail = FALSE)
```

```
cat("=== MANUAL ONE-WAY ANOVA RESULTS ===\n")  
cat("Variable: koi_period grouped by koi_fpflag_nt\n")  
cat("F-Statistic:", f_statistic, "\n")  
cat("P-Value:", p_value, "\n")
```

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Output:

The screenshot displays the RStudio interface. The console on the left shows the execution of R code for a one-way ANOVA. The code reads a CSV file, filters for non-NA values, calculates group means and variances, and performs the ANOVA test. The environment pane on the right shows the objects created during the process.

```
R > R 4.5.2 - ~/...
> library(tidyverse)
> df <- read.csv("kepler-exoplanet.csv") %>%
+   select(koi_fpflag_nt, koi_period) %>%
+   filter(!is.na(koi_period))
> grand_mean <- mean(df$koi_period)
> N <- nrow(df)
> stats_table <- df %>%
+   group_by(koi_fpflag_nt) %>%
+   summarise(
+     group_mean = mean(koi_period),
+     n = n(),
+     group_var = var(koi_period)
+   ) %>%
+   mutate(
+     # SSB calculation: n * (Group Mean - Grand Mean)^2
+     ssb_part = n * (group_mean - grand_mean)^2,
+     # SSW calculation: (n - 1) * Variance
+     ssw_part = (n - 1) * ifelse(is.na(group_var), 0, group_var)
+   )
> SSB <- sum(stats_table$ssb_part)
> SSW <- sum(stats_table$ssw_part)
> df_between <- nrow(stats_table) - 1
> df_within <- N - nrow(stats_table)
> MSB <- SSB / df_between
> MSW <- SSW / df_within
> f_statistic <- MSB / MSW
> p_value <- pf(f_statistic, df_between, df_within, lower.tail = FALSE)
> cat("=== MANUAL ONE-WAY ANOVA RESULTS ===\n")
=== MANUAL ONE-WAY ANOVA RESULTS ===
> cat("Variable: koi_period grouped by koi_fpflag_nt\n")
Variable: koi_period grouped by koi_fpflag_nt
> cat("F-Statistic:", f_statistic, "\n")
F-Statistic: 10.06551
> cat("P-Value:", p_value, "\n")
P-Value: 0.001515484
>
```

The Environment pane shows the following objects:

Object	Size
data	2930 obs. of 82 variables
data_paired	2930 obs. of 2 variables
df	9564 obs. of 2 variables
long_data	5860 obs. of 2 variables
paired_t_test	List of 10
stats_table	2 obs. of 6 variables

The Values pane shows the following data for the stats_table object:

Variable	Value
df_between	1
df_within	9562L
difference	int [1:2930] -1656 -896 -1329 -2110 -227 ...
f_statistic	10.0655138312771
grand_mean	75.6713584248239
MSB	17915149.1375978
MSW	1779854.4056369
N	9564L
p_value	0.0015154841342345
SSB	17915149.1375978
SSW	17018967826.7001